

HW 4

To be completed by Friday, March 6th

1. Use “dataHW4 Problem1.mat” to complete this exercise. In this problem you will work with a multinomial choice model very similar to the one from previous problem sets. Upon graduating high school, individuals have three options: work ($Y=0$), attend a 2-year college ($Y=1$), attend a 4-year college ($Y=2$). The available student demographic characteristics (X_i) are a constant, an indicator for whether either parent graduated from a 4-year college, and the student’s high school GPA respectively. There is one choice specific variable, price measured in thousands of dollars (Z_{price}). Each column of Z_{price} reflects the price for the corresponding choice. In other words, the first column is the price for the work choice, the second column for the 2-year choice, and so on. Note that I assume that price for the work option is zero.

- (a) Estimate a mixed logit using simulation with $R = 500$ draws. The coefficient on price is normally distributed with mean μ_γ and variance σ_γ^2 . The unknown coefficients for the student demographic variables do not vary in the population. In other words, the only random coefficient is on price. Normalize the utility of work to zero. Report parameter estimates and SEs for the β s, μ_γ , and σ_γ .
- (b) Calculate the average marginal effect of having a parent with a BA on the likelihood of attending a 4-year college. To calculate the average marginal effect you will need to integrate out over the price coefficient, similar to what you did in estimation. Here you can use 100 draws per observation.
- (c) Now I want you to calculate the standard error of the estimated marginal effect using a parametric bootstrap. To avoid having to re-estimate the model, for each bootstrap iteration take a random draw from the joint distribution of the estimates, which we know is approximately normal with a mean of $[\hat{\beta}' \hat{\mu}_\gamma \hat{\sigma}_\gamma]'$ and variance given by the inverse of the Hessian from part (a). Use the Cholesky decomposition to make these draws. With this draw re-calculate the marginal effect (which will again require integrating over the price coefficient - take 100 draws). Do 100 bootstrap iterations.
- (d) Re-estimate the mixed logit model from part (a) using 5-point Gaussian-Hermite quadrature to approximate the integral over the choice probabilities.

True parameter values: $\beta_{2yr} = [-1.0; 0.25; 0.45]$, $\beta_{4yr} = [-2.5; 0.75; 1.25]$, $\mu_\gamma = -0.15$, and $\sigma_\gamma = 0.035$.

2. Use “dataHW4 Problem2.mat” to complete this exercise. In this problem you will work with panel data to estimate a logit model that allows for discrete unobserved heterogeneity. During the pandemic working from home became the norm, and many people continue to work from home. Over the course of six months, you collect data from women between the ages of 30 and 40 who are working full-time. Each month you observe whether the woman was working at home full time ($Y=1$). The six columns of Y reflect her response for each month. For each respondent you also observe X_i containing the number of children at home and years of schooling (in addition to a constant). You also observe distance from the physical workplace in miles (Z_{dist}). Note that X_i and distance are fixed for each women over the 6 months.
 - (a) You suspect that in addition to children and schooling, women likely differ in their tastes for working at home. Assume that there are two unobserved types of women, one type prefers the office, the other prefers to work from home, all else equal. Type is assumed to be uncorrelated with X_i or distance. Estimate a mixed logit model for the decision to work from home that allows for these two types of women. Note that this means there will be two additional parameters to estimate beyond the coefficients associated with X_i and distance. You need to estimate the proportion of Type 2 women, and the additional preference these women have for working at home. Report coefficients and SEs.

- (b) Estimate the same model as in part (a), except use the EM algorithm. Report coefficients. How would you obtain SEs?

True parameter values: $\beta = [-0.5; 0.5; -0.2]$, $\gamma = 0.05$, $\delta = 2.0$ (Type 2 preference for home), and $\pi_2 = .685$ (population share of Type 2s).